PROJECT NAME: The "Safe-Night" Transit

Solution Approach:

The system improves night travel safety using **real-time tracking and smart alerts**.

- **Live Tracking:** GPS tracks user/vehicle location continuously
- **Monitoring:** Detects route deviation, delays, and unusual stops
- **SOS Feature:** One-click emergency alert with live location
- **Risk Detection:** Identifies unsafe areas using past data
- **Notifications:** Sends alerts to users and emergency contacts

Design Explanation:

◈ **Architecture:**

Follows a **3-tier structure**:

- **Frontend:** Mobile/Web app (UI, tracking, SOS)
- **Backend:** Handles logic, alerts, and analysis
- **Database:** Stores users, trips, and location data

◈ **Key Modules:**

- **User Module:** Login, trip start/end
- **Tracking Module:** GPS data collection
- **Alert Module:** SOS & notifications
- **Risk Module:** Detects unsafe conditions
- **Admin Module:** Monitoring & reports

◈ **Workflow:**

1. User starts trip
2. GPS tracking begins
3. System monitors route
4. Alerts triggered if risk detected
5. SOS sends emergency notification

Summary:

A **real-time safety system** that combines tracking, alerts, and risk analysis to ensure secure night travel.